

Python Fundamentals

Custom knowledge-base material for the AI & Data Science RAG Chatbot project.

1. What is Python?

Python is a high-level, interpreted programming language known for readable syntax. It is widely used in automation, data science, machine learning, web development, and artificial intelligence.

2. Variables and Data Types

A variable stores a value. Common data types include int for whole numbers, float for decimal numbers, str for text, bool for True or False, list for ordered mutable collections, tuple for ordered immutable collections, dict for key-value pairs, and set for unique values.

3. Control Flow

Conditional statements use if, elif, and else to make decisions. for loops iterate over items in a sequence. while loops repeat while a condition remains true. break exits a loop and continue skips to the next iteration.

4. Functions

A function is a reusable block of code defined with def. Functions can accept parameters and return values. Good functions have one clear responsibility and meaningful names.

5. File Handling

Python can read and write files using open(). Common modes are r for reading, w for writing, a for appending, and b for binary data. The with statement closes files automatically.

6. Exceptions and Secure Coding

Use specific exceptions such as FileNotFoundError and ValueError. Avoid broad except blocks that hide problems. Never hardcode passwords or API keys; use environment variables and .env files. Validate user input and review AI-generated code before using it.

7. Data Science Libraries

NumPy provides efficient numerical arrays. Pandas provides Series and DataFrame structures for data manipulation. Matplotlib is used for visualization. These libraries are common foundations for machine learning workflows.

8. Key Terms

Algorithm: a sequence of steps to solve a problem. Dataset: a collection of observations. Function: reusable code. Library: reusable software functionality. Debugging: finding and fixing errors.